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PRE – PSPM SEMESTER 1

CHEMISTRY

SK015

(2 HOURS)

INSTRUCTION TO CANDIDATE:

- This question paper consists of 6 questions.
- Answer ALL questions.
- All steps must be shown clearly. Use a new page for each question.
- The use of non-programmable scientific calculator is permitted.

Name: _____

Practikum: _____

Question	Marks
1	/21
2	/10
3	/17
4	/9
5	/9
6	/14

1. a) Magnesium has three naturally occurring isotopes with isotopic mass 23.9850 u, 24.9858 u and 25.9826 u. Mg-24 is the most abundance isotope with 78.70 %.

   i. Calculate the percentage abundance of Mg-25 and Mg-26.

   ii. Draw the mass spectrum for the isotopes above.

   [6 *marks*]

   b) Copper (II) sulfate, $CuSO_4$ often used because it forms a vibrant blue solution, making it ideal for experiments involving calorimeter or spectroscopy. A commercial copper (II) sulfate, $CuSO_4$ solution contains 23**.**85 % by mass**.** Calculate the molality of the solution above.

   [5 *marks*]

   c) A matriculation student performs an experiment by dissolving 10.0 g of zinc powder into 100.0 mL of 0.50 M potassium nitrate solution, KNO_3 in a basic medium at 25 ^oC, 1 atm. The ionic species involved in the redox reaction are given below:

$$Zn\ (s)\ +\ NO_3^-\ (aq)\ \longrightarrow\ [Zn(OH)_4]^{2-}\ (aq)\ +\ NH_3\ (g)$$

   i. Balance the equation above.

   ii. Calculate the amount of ammonia, NH_3 formed in mL.

   [10 *marks*]

2. a) The Brackett series is a group of spectral lines in the infrared region of the electromagnetic spectrum, resulting from electron transitions in hydrogen atoms.

   i. Calculate the wavelength and the frequency for the electron transition that correspond to the second spectral line in the Brackett series.

   ii. Calculate the energy (in $kJmol^{-1}$) emitted by 1 mole of electrons for the above transition.

   [7 *marks*]

2. b) Given the following sets of quantum numbers for the electrons in the highest energy orbitals of atom J:

(3, 0, 0, +½), (3, 0, 0, -½), (3, 1, 0, +½) and (3, 1, -1 +½)

(i) Draw the orbital diagram that represents the valence electrons of atom ***J***.
(ii) State the number of electrons in orbital of J with azimuthal quantum number, ***l*=0**.
(iii) Suggest the most stable ion for ***J***.

[3 *marks*]

3. a) Referring to cyanate ion, NCO^-
   i. Draw three (3) possible Lewis structure of cyanate ion.
   ii. Determine the most plausible Lewis structure of cyanate ion and give your reason.

[5 *marks*]

b) Sulfur tetrafluoride, SF_4, and sulphur hexafluoride, SF_6 are simple covalent molecules.
   i. Draw the Lewis structures for SF_4 and SF_6
   ii. Using Valence Shell Electron Pair Repulsion (VSEPR) theory, explain why SF_4 and SF_6 have different molecular shapes.
   iii. Deduce the polarity of the molecule. Give a reason for your answer.

[8 *marks*]

c) Explain each of the following statements.
   i. Silicon has higher melting point than Cl_2.
   ii. H_2O has higher boiling point than HF.

[4 *marks*]

4. a) An amount of 40.9 O_2 gas and 25.0g N_2 gas are mixed in a close vessel of 8.10 L at 298.15 K. Calculate the total pressure of the mixture. CO_2 gas is then added to above mixture to give total pressure of 1.05×10^6 Nm^{-2} consisting N_2, O_2 and CO_2 in the ratio of 5:6:9 respectively. Determine the partial pressure, in atm for each gas in the mixture.

[6 *marks*]

4. b) Nitrogen gas is a real gas that deviates from ideality. It is the most abundance gas found in the atmosphere.
   i. Why does nitrogen gas behave as a real gas at room temperature?
   ii. State the condition under which nitrogen become a real gas.
   iii. How can nitrogen gas be made behave as an ideal gas

[3 *marks*]

5. a) For a reaction:

$$NH_4Cl_{(s)} \rightleftharpoons NH_{3(g)} + HCl_{(g)}$$

If 1.85×10^{-2} mol NH_4Cl, 1.50×10^{-2} mol NH_3 and 7.30×10^{-1} mol HCl are placed into a 2.0 L vessel at 25 °C to reach equilibrium. Given that the Kc for the reaction is 1.18×10^{-19}.

   i. Determine the direction of the reaction to achieve equilibrium.
   ii. Calculate the equilibrium constant, Kp for the reaction at 25 °C.

[5 *marks*]

   b) For the following reactions at 298K;

Reaction I: $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ $\Delta H = +93.0 kJmol^{-1}$
Reaction II: $2NO(g) + Cl_{2(g)} \rightleftharpoons 2\ NOCl(g)$ $\Delta H = -77.2\ kJmol^{-1}$

By using Le Chatelier's Principle, briefly explain how the effect of temperature will increase the amount of concentration:

   i. PCl_5 in reaction I

   ii. NOCl in reaction II

[4 *marks*]

6. a) A 0.10 mol dm^{-3} solution of chlorous acid, $HClO_2$, has a pH of 1.20. Calculate the degree of dissociation of the chlorous acid.

[6 marks]

b) In an experiment, 25.0 mL of 0.50 M HCl was titrated against 0.50 M CH_3NH_2 solution. The end point is reached when 25.0 mL of CH_3NH_2 solution was added according to the equation below.

i. Predict the pH of the solution at the end point.

[3 *marks*]

ii. State a suitable indicator for this titration.

[1 *marks*]

c) Solid PbI_2 is placed in a beaker of water. After some time, the concentrations of Pb^{2+} ions and I^- ions are measured and found to be:
$[Pb^{2+}] = 1.0 \times 10^{-4}$ M
$[I^-] = 1.0 \times 10^{-4}$ M
Does a precipitate form in this situation?
$[Ksp(PbI_2) = 8.5 \times 10^{-9}]$

[4 *marks*]

END OF QUESTION PAPER